

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING

1st/2nd Semester of B. Tech. Examination (ME/CL/EE)

April - May 2018

ME 143 - BASICS OF CIVIL AND MECHANICAL ENGINEERING

Date: 07/05/2018 (Monday)

Time: 01.30 p.m. To 04.30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Do as directed:

[07]

- 1) In Chain Survey the area is divided into(Triangles, Rectangles)
- 2) In load bearing structure, the load of the building is transfer through.....
(Beam & Column, wall)
- 3) The FB of line is 65^0 , its BB is (245^0 , 25^0 , 65^0)
- 4) The angle between two plain mirrors of the optical square is..... (60^0 , 45^0)
- 5) The plain scale is used to read consecutive units. (One, Two, Three)
- 6) A Prismatic Compass is used to measure Bearing. (WCB, QB, Linear)
- 7) Indirect Ranging is also known as (Direct Ranging, Reciprocal Ranging)

Q - 2 (a) Explain scope of civil engineering

[03]

(b) Define Ranging. Write the difference between Direct Ranging and Indirect ranging.

[04]

(c) Classify the building according to its occupancy and explain it in details.

[07]

OR

(b) Write difference between Prismatic compass and Surveyor compass.

[04]

(c) Classify the building according to structure and explain frame structure in details with neat sketch.

[07]

Q - 3 (a) Enlist the various materials used in building construction. Explain properties of good bricks.

[04]

Q - 3 (b) Enlist various design loads acting on the building and explain any two in details

[05]

(c) A chain was tested before starting the survey and was found to be exactly 30.00m. After measuring 300 m, it was again tested and was found to be 3 cm too short. After measuring a total distance of 600 m, it was again tested and found to be 5 cm too

[05]

short. Calculate actual length of line.

OR

Q – 3(b) Enlist Various Principles of Planning and Explain Aspect and Privacy in details. [05]

(c) The following bearings were taken on a closed traverse ABCD in clockwise direction. [05]

Calculate interior angle of the traverse and apply necessary checks.

Line	FB	BB
AB	45° 00'	225° 00'
BC	123° 30'	303° 30'
CD	181° 00'	1° 00'
DA	289° 00'	109° 00'

SECTION – II

Q - 4 Answer the questions below. [07]

1) 1bar = _____ N/m²

A. 10³ B. 10⁵ C. 10⁻³ D. 10⁻⁵

2) The relation between Cp and Cv is:

A. Cp - Cv = R B. Cv - Cp C. Cp + Cv = R D. Cp . Cv

3) The phase change from solid to vapour is called

A. sublimation B. vaporization C. pressurization D. temperature

4) Dryness fraction of a steam (x) is given by:

A. ms / mw B. mw / ms C. (ms + 1) / mw D. ms / (ms +mw)

5) Which part used in I.C.Engine to convert reciprocating motion of piston to rotary motion of output shaft

A. Connecting rod B. Crank shaft C. Cam shaft D. Gudgeon pin

6) Which types of coupling used to connecting shafts, whose axis are parallel but not in one line?

A. muff coupling B. flexible coupling C. Oldham's coupling D. flange coupling

7) Which one is the water tube boiler?

A. Cochran B. Lancashire C. Babcock and Wilcox D. none of these

Q – 5.a Drawing a diagram explain construction and working of combined calorimeter. [05]

OR

Q – 5.a Explain working of main parts of centrifugal pump with neat sketch. [05]

Q – 5.b Do as directed: [04]

1) Write short note on helical gear.

2) Write down the function of Economiser and Air Preheater

- Q – 5.c** Mention different parts of vapor compression refrigeration cycle along with their functions. Also draw a neat diagram of vapor compression refrigeration cycle. **[05]**

OR

- Q – 5.c** Differentiate brake and clutch. Explain Oldham's coupling. **[05]**

- Q – 6.a** Derive Mayer's equation with usual notations. **[07]**

- Q – 6.b** One kg of air at a pressure of 1 bar, temperature of 27°C is compressed to a pressure of 10 bar isothermally and adiabatically. Calculate work done, heat transfer and change in internal energy in both the cases. Take $C_p = 1.005 \text{ kJ/kg K}$, $C_v = 0.718 \text{ kJ/kg K}$. **[07]**

OR

- Q – 6.a** Explain with the help of neat sketches, the working of two stroke petrol engine. **[07]**

- Q – 6.b** A single cylinder 4 stroke diesel engine has following performance test data: **[07]**
Bore of cylinder is 15cm, stroke is 20cm, area of indicator diagram is 3.2 cm^2 , length of diagram 5cm, indicator spring constant is 100 N/cm/cm, engine speed is 600 rpm, the load on dynamometer is 42 kg, length of dynamometer arm is 35cm. Calculate indicated power, brake power, friction power and mechanical efficiency.
